Tech Bite #3: Setting Up Your Station

The hands-on Bite -- how the pieces of a station connect: power, the antenna feed line, and a computer for digital modes.

Group T4A · 12 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

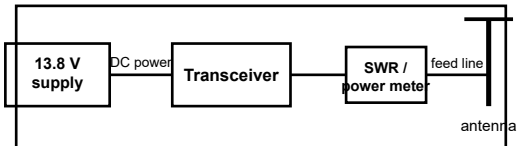
The Big Picture

This Bite is the practical one. A working station is just a few pieces connected correctly: a radio, something to power it, an antenna, and – if you want digital modes – a computer. Learn how they hook together and the rest of operating is easy.

Power: Feeding the Radio

Most VHF/UHF radios run on about **13.8 volts** – the voltage of a "12-volt" vehicle system with the engine running. A typical 50-watt mobile FM radio needs a supply rated for **13.8 volts at about 12 amperes**. (T4A01) Use **short, heavy-gauge wire** for the DC connection, so the voltage does not sag while you transmit. (T4A03)

In a vehicle, connect the radio's negative power lead straight to the **battery's chassis ground** – not to a random metal bracket. (T4A11) To estimate how long a battery will last, **divide its ampere-hour rating by the radio's average current draw**. (T4A09)

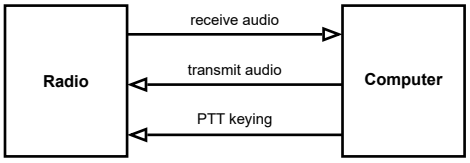


A basic station: a 13.8 V supply powers the radio; the feed line carries the signal to the antenna, with an SWR/power meter inline.

The Antenna Side: Meters in the Feed Line

An SWR meter or RF power meter belongs **in the feed line, between the transmitter and the antenna** – that is the only place it can read what the antenna is actually doing. (T4A05) When you buy an SWR meter, match it to the **frequency and power level** you plan to measure. (T4A02) When you bond equipment together for RF, use **flat copper strap**; it has far less impedance at radio frequencies than ordinary round wire. (T4A08)

Adding a Computer: Digital Modes



The digital-mode interface carries three signals between the radio and the computer: receive audio, transmit audio, and PTT keying.

Digital modes like FT8 are run by a computer. You connect the radio's audio in and out to the **audio output and input of a computer** running the software. (T4A04) The link carries three signals: **receive audio** (radio to computer), **transmit audio** (computer to radio), and **transmitter keying**, or PTT (computer to radio). (T4A06)

One specific connection in that interface is the computer's "line in" to the transceiver's **speaker connector**. (T4A07) A **digital-mode hotspot** is a small box that connects your radio to a digital voice or data network over the internet. (T4A10)

One Handy Tool

An **electronic keyer** is a device that helps you send Morse code by hand – it shapes the dots and dashes evenly as you tap its paddle. (T4A12)

Mental Cheat Sheet

- A 50 W mobile FM radio needs **13.8 V at about 12 A**.
- DC power wire: **short and heavy-gauge**, to stop voltage drop on transmit.
- Mobile: negative lead to the **battery's chassis ground**.
- Battery runtime = **amp-hours / average current draw**.
- SWR / RF power meter goes **in the feed line**, between radio and antenna.
- Bond equipment with **flat copper strap** (low impedance at RF).
- Digital modes: radio audio in/out to a **computer** running the software.
- The digital interface carries **receive audio, transmit audio, and PTT keying**.

Common Traps

- A 50 W radio runs on **13.8 V**, not 24 V – and needs about **12 A**, not 4 A.
- The power/SWR meter goes **in the feed line** – not at the power supply.
- Bond with flat copper **strap** – not round wire or coax braid.
- A hotspot links to a **digital network** – it is not an FT8 box or an RTTY decoder.
- The mobile negative lead goes to the **battery ground** – not "any metal part."

Takeaway: A station is power + radio + antenna, plus a computer for digital modes. Power it with **13.8 volts** through **short, heavy wire**; put your SWR or power meter **in the feed line**; bond equipment with **flat strap**. For digital modes, the radio's audio connects to a computer running the software. Connect the pieces correctly and you are on the air.

Bite 3 Question Bank (T4A)

T4A01. Which of the following is an appropriate power supply rating for a typical 50-watt output mobile FM transceiver?

B. 13.8 volts at 12 amperes

T4A02. Which of the following should be considered when selecting an accessory SWR meter?
A. The frequency and power level at which the measurements will be made
B. The distance that the meter will be located from the antenna
C. The maximum SWR anticipated on the transmission line
D. The ability of the meter to compensate for a poor transmission line match to the antenna

T4A03. Why are short, heavy-gauge wires used for a transceiver's DC power connection?
A. To minimize voltage drop when transmitting
B. To provide a close match to the power supply output impedance
C. To avoid RF interference
D. To minimize radiative losses in the power cable

T4A04. How are the audio input and output of a transceiver connected in a station configured to operate using FT8?
A. To a computer running a terminal program and connected to a terminal node controller unit
B. To the audio output and input of a computer running FT8 software
C. To an FT8 conversion unit, a keyboard, and a computer monitor
D. To a computer connected to the FT8converter.com website

T4A05. Where should an RF power meter be installed?
A. In the feed line, between the transmitter and antenna
B. At the power supply output
C. In parallel with the push-to-talk line and the antenna
D. In the power supply cable, as close as possible to the radio

T4A06. What signals are used in a computer-radio interface for digital mode operation?
A. Receive and transmit mode, status, and location
B. Antenna and RF power
C. Receive audio, transmit audio, and transmitter keying
D. NMEA GPS location and DC power

T4A07. Which of the following is one of the connections required between a computer and a transceiver to operate digital modes?
A. Computer "line out" to transceiver push-to-talk
B. Computer "line in" to transceiver push-to-talk
C. Computer "line in" to transceiver speaker connector
D. Computer "line out" to transceiver speaker connector

T4A08. Which of the following conductors is preferred for bonding at RF?
A. Copper braid removed from coaxial cable
B. Copper-clad steel wire
C. Twisted-pair cable
D. Flat copper strap

T4A09. How can you determine the length of time that equipment can be powered from a battery?
A. Divide the watt-hour rating of the battery by the peak power consumption of the equipment
B. Divide the battery ampere-hour rating by the average current draw of the equipment
C. Multiply the watts per hour consumed by the equipment by the battery power rating
D. Multiply the square of the current rating of the battery by the input resistance of the equipment

T4A10. What function does a digital mode hotspot perform for nearby transceivers?
A. Communication with a digital voice or data network
B. FT8 digital communications via AFSK using a smartphone connected to the internet
C. RTTY encoding and decoding without a computer
D. High-speed digital communications for meteor scatter

T4A11. Where should the negative power return of a mobile transceiver be connected in a vehicle?
A. At the 12-volt battery chassis ground
B. To the shell of the power connector
C. To any metal part of the vehicle
D. Through the transceiver's mounting bracket

T4A12. What is an electronic keyer?
A. A device for switching antennas from transmit to receive
B. A device for voice activated switching from receive to transmit
C. A device that assists in manual sending of Morse code
D. An interlock to prevent unauthorized use of a radio

Answer Key

T4A01: D	T4A02: A	T4A03: A	T4A04: B
T4A05: A	T4A06: C	T4A07: C	T4A08: D
T4A09: B	T4A10: A	T4A11: A	T4A12: C